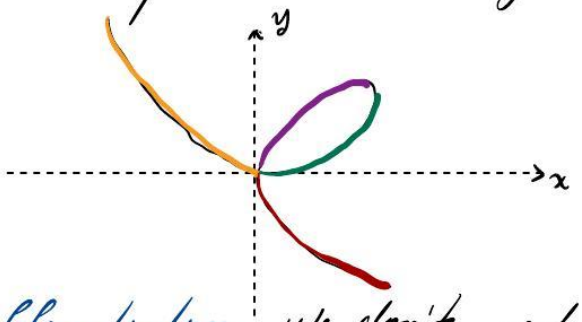
Lecture 9, Jan 9th, 2025.

For example let $x^3+y^3=6xy$.




Implicit Differentiation. We don't need to solve an equation for $y$ in terms of $x$ in order to find the derivative of $y$. Instead we can use implicit differentiation. This consist of differentiating both sides of the equation with respect to $x$ and then solving the resulting equation for $\frac{dy}{dx}$.

Example. a). Find $y'$ if $x^3+y^3=6xy$.

b). Find the tangent line at $(3,3)$.

c). At what point in the first quadrant is the tangent line horizontal.

Sln. a).

$$\frac{d}{dx}(x^3+y^3)=\frac{d}{dx}(6xy).$$

$$\frac{d}{dx}(x^3)+\frac{d}{dx}(y^3)=6\frac{d}{dx}(xy)$$

$$3x^2+3y^2\cdot y'=6\left(\frac{d}{dx}(x)\cdot y+x\cdot y'\right)$$

$$3x^2+3y^2y'=6(y+xy')$$

Solve for $y'$: $3x^2+3y^2y'=6y+6xy'$

$$3x^2-6y=6xy'-3y^2y'$$

$$3x^2-6y=y'(6x-3y^2)$$

$$y'=\frac{3x^2-6y}{6x-3y^2}=\frac{x^2-2y}{2x-y^2}.$$

b). at $(3,3)$
$x_1\ y_1$

$$m = y'\Big|_{(3,3)} = \frac{3^2 - 2(3)}{2(3) - 3^2} = -1.$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -1(x-3)$$

$$y = -x + 3 + 3 = -x + 6.$$

c). We are looking for $y' = 0 \iff \frac{x^2 - 2y}{2x - y^2} = 0 \iff x^2 = 2y \iff y = \frac{1}{2}x^2$.

We know that, $x^3 + y^3 = 6xy$.

$$x^3 + \left(\frac{1}{2}x^2\right)^3 = 6x\left(\frac{1}{2}x^2\right)$$

$$x^3 + \frac{1}{8}x^6 = 3x^3$$

$$x^6 = 16x^3$$

$$x^3(x^3 - 16) = 0$$

$x = 0$ or $x^3 = 16$ $\quad x = \sqrt[3]{16} = 2^{4/3}$.

if $x = 0$ then $y = 0$.

if $x = 2^{4/3}$ $\quad y = \frac{1}{2}(2^{4/3})^2 = 2^{5/3}$.

then, the tangent line is horizontal at $(0,0)$ and $(2^{4/3}, 2^{5/3})$.

Example. find $y'$ if $\sin(x+y) = y^2 \cos x$.

Sln. $\quad \frac{d}{dx}(\sin(x+y)) = \frac{d}{dx}(y^2 \cos x)$

By C-R $\rightarrow$ $\cos(x+y) \cdot \frac{d}{dx}(x+y) = \frac{d}{dx}(y^2) \cdot \cos x + y^2 \cdot \frac{d}{dx}(\cos x)$.

$$\cos(x+y)(1+y') = 2yy' \cos x - y^2 \sin x.$$

[ $G(y) = y^2$ $\quad G(y) = F(H(y))$ $\quad F(x) = x^2$, $H(y) = y$. $\quad$ By CR $\quad G'(y) = 2yy'$ ]

$$\cos(x+y) + y'\cos(x+y) = 2yy'\cos x - y^2\sin x.$$

$$\cos(x+y) + y^2\sin x = 2yy'\cos x - y'\cos(x+y).$$

$$\cos(x+y) + y^2\sin x = y'(2y\cos x - \cos(x+y)).$$

$$y' = \frac{\cos(x+y) + y^2\sin x}{2y\cos x - \cos(x+y)}.$$

## Second derivatives of implicit functions.

**Example.** Find $y''$ if $x^4+y^4 = 16$.

i). $y'$:

$$\frac{d}{dx}(x^4+y^4) = \frac{d}{dx}(16).$$

$$4x^3 + 4y^3\cdot y' = 0.$$

$$y' = -\frac{x^3}{y^3}.$$

ii). $y''$:

$$\frac{d}{dx}(y') = \frac{d}{dx}\left(-\frac{x^3}{y^3}\right)$$

$$y'' = \frac{-3x^2\cdot y^3 - (-x^3)(3y^2\cdot y')}{(y^3)^2}.$$

$$y'' = \frac{-3x^2y^3 + 3x^3y^2y'}{y^6}$$

$$y'' = \frac{-3x^2y^3 + 3x^3y^2\left(-\frac{x^3}{y^3}\right)}{y^6}$$

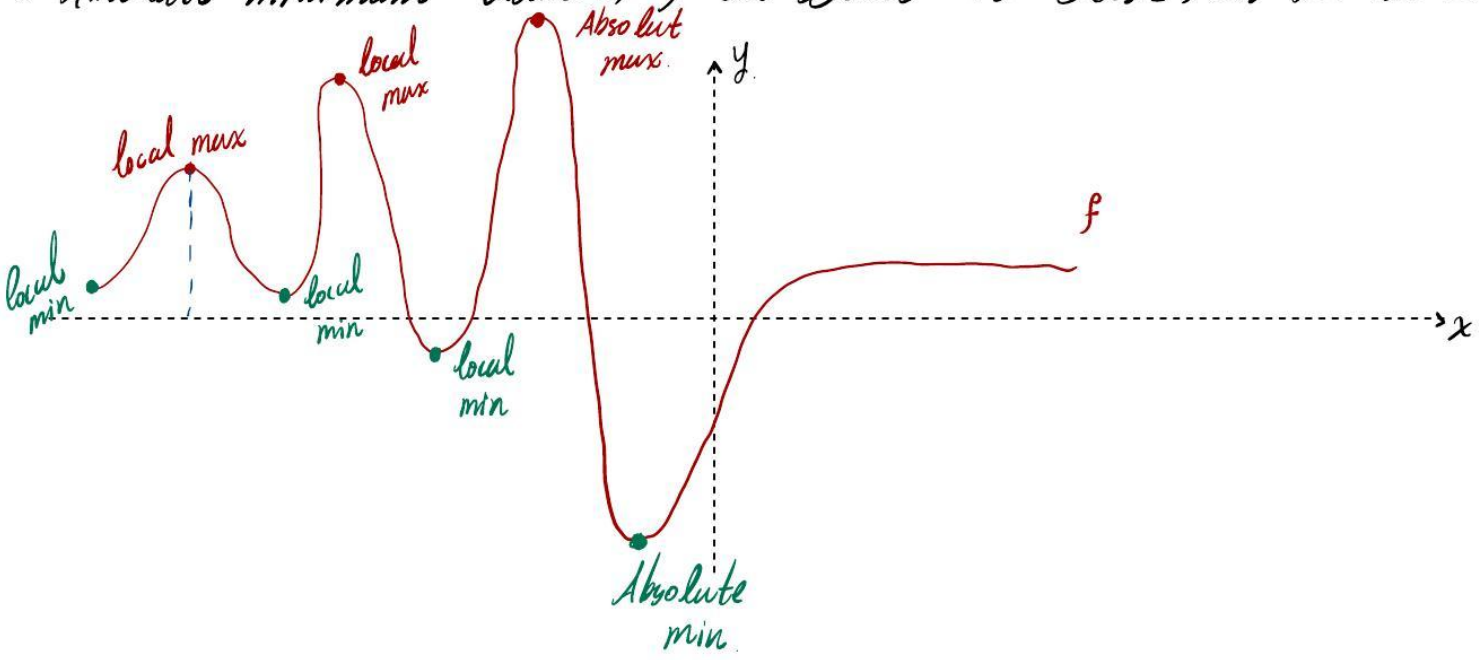
$$y'' = \frac{-\frac{3x^2y^3}{1} - \frac{3x^6}{y}}{y^6}$$

$$y'' = \frac{-3x^2y^4 - 3x^6}{y^7} = -\frac{3x^2(y^4+x^4)}{y^7} = \frac{-48x^2}{y^7}.$$

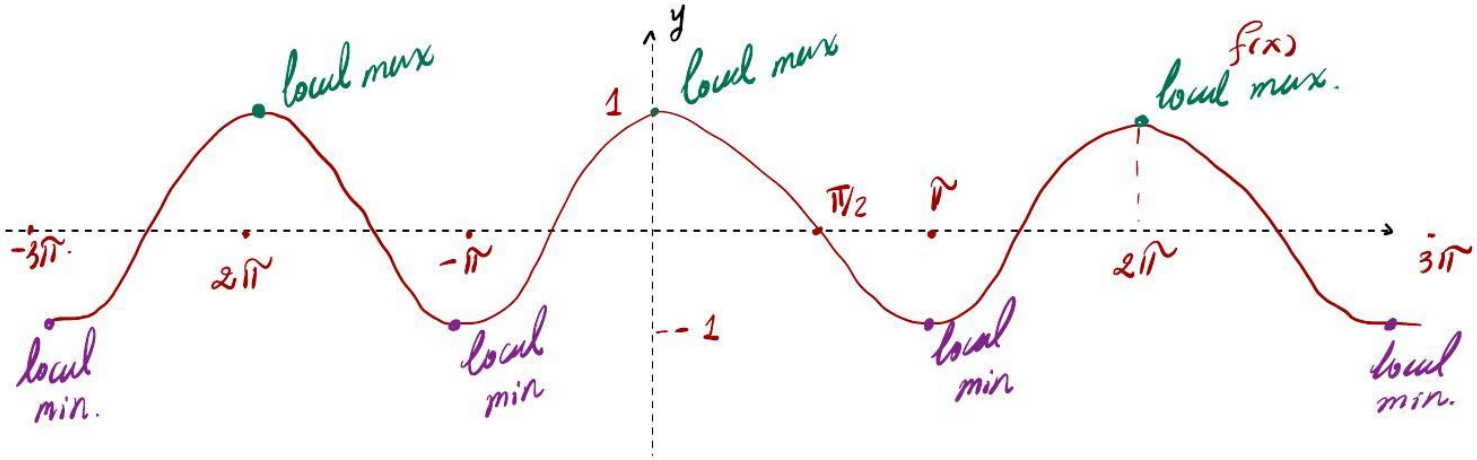
## 3.1. Maximum and minimum Values.

**Def.** Let $c$ be a number in $Domf$. Then $f(c)$ is the:

- Absolute maximum value of $f$ on $Domf$ if $f(c) \geq f(x)$ for all $x \in Dom$.
- Absolute minimum value of $f$ on $Domf$ if $f(c) \leq f(x)$ for all $x \in Dom$.




**Example.** Let consider $f(x) = \cos x$, this function on $\mathbb{R}$ has only local minimum and local maximum.




**Thm.** (The Extem Value theorem) if $f$ a continuous function on a closed interval $[a,b]$ then $f$ attains an absolute maximum $f(c)$ and an absolute minimum $f(d)$ for some points $c, d \in [a,b]$.

### Critical Numbers and closed interval Method

Let $f$ is a function then the values where $f'(x)=0$ for $x \in Domf$

are called critical numbers of $f$.

Now, in particular $\text{Dom } f = [a,b]$ then the critical values of $f$ are the values s.t $f'(x)=0$ and $x=a$, $x=b$.

**Thm** (Fermat's theorem). If $f$ has a local maximum or minimum at $c$, and $f'(c)$ exists, then $f'(c)=0$.

(+). (New critical number). We say that the function $f$ has a critical value at $c$ if $f'(c)=0$ or $f'(c)$ does not exists

**Example.** Find the critical points of:

a). $f(x) = x^3 - 3x^2 + 1$

b). $f(x) = x^{3/5}(4-x)$.

**Sln.** a). $\text{Dom } f = \mathbb{R}$. $f'(x) = 3x^2 - 6x$. Since $f'(x)$ exists for $x \in \mathbb{R}$ the only critical values are $f'(x)=0$ that is $x=0$, $x=2$.

b). $\text{Dom } f = \mathbb{R}$ Then, by the product rule.

$$f'(x) = \frac{3}{5} x^{3/5-1}(4-x) + x^{3/5}(-1)$$

$$= -x^{3/5} + \frac{3(4-x)}{5x^{2/5}} = \frac{-5x + 12 - 3x}{5x^{2/5}} = \frac{12-8x}{5x^{2/5}}.$$

then: i). $f'(x)=0 \iff 12-8x=0 \iff x=\frac{3}{2}$.

ii). $f'(x)$ DNE $\iff 5x^{2/5}=0 \iff x=0$

} are the critical points.

**Prop.** If $f$ has a maximum or minimum at $c$, then $c$ is critical value of $f$.

**The closed interval method.** To find the absolute maximum and minimum values of a continuous function $f$ on a closed interval $[a,b]$.

1. Find the values of $f$ at the critical values of $f$ in $(a,b)$.
2. Find the values of $f$ at the endpoints of the interval $[a,b]$.

3. The largest of the values from step 1 and 2 is the absolute maximum; the smallest of these values is the absolute minimum.

**Example.** Find the absolute maximum and minimum of

$$f(x) = x^3 - 3x^2 + 1 \qquad -\frac{1}{2} \le x \le 4.$$

**Sln.** $f(x)$ is continuous on $[-\frac{1}{2}, 4]$. We can use the closed interval method we know that

$$f'(x) = 3x^2 - 6x = 3x(x-2)$$

So, the critical points are $x=0$, $x=2$. Notice that each of these lies in the interval $(-\frac{1}{2}, 4)$.

**step 1** $f(0) = 1$, $f(2) = -3$.

**step 2.** $f(-\frac{1}{2}) = -\frac{1}{8} - \frac{3}{4} + 1 = \frac{1}{8}$; $f(4) = 4^3 - 3(16) + 1 = 17$.

**step 3.** then $f(2) = -3$ is the smallest then is the absolute minimum.
the $f(4) = 17$ is abs. max.